



Vojtěch Špaček

Senior C++ engineer — safety-critical embedded & HIL test automation | Linux/networking | agentic-AI (MCP) tooling

Pragmatic programmer and creative problem solver — systems, verification and open source.

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Nationality: EU (Czech Republic) **Place of residence:** Brno, CZ

Work permit: EU citizen (Czech Republic) — no permit required in the EU/EEA; B-permit eligible for Switzerland.

Profile Summary

Senior C++ engineer working on safety-critical and real-time systems: aircraft onboard diagnostics at Honeywell, electron-microscopy detector software at Thermo Fisher Scientific, and a Linux inspection proxy and firewall at Trusted Network Solutions.

Most recently sole owner of the hardware-in-the-loop verification suite and automated bench for a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), raising measured conformance to 259 of 271 tracked state transitions (95.6%).

Linux, networking and security specialist — Netlink, nftables, TLS inspection, Kerberos/LDAP/OCSP — with C++20 daemons shipped as Debian packages and a Rust reimplementation published on crates.io.

Builds agentic-AI tooling on the Model Context Protocol: an agent harness adopted into a daily client delivery workflow, and MCP-Prompts, published to npm, Docker Hub and the MCP directories (Glama, MCP Market, PulseMCP) with 97 stars and 21 forks.

Work Experience

Test Automation Engineer (Python, C#)

Resideo (Ademco) (via CoolPeople (contractor))

Hardware-in-the-loop verification of a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), as sole owner of the verification test suite and of the automated bench that drives it. Owned delivery end to end: requirement traceability, issue tracking, branch and pull-request workflow, Conan packaging, CI integration and monthly client reporting.

- **Designed an Arrange/Act/Assert validation pipeline in Python with eight independent gates covering preconditions, stimulus timing, state sequence, alarms, signal history, cross-layer consistency, timer accuracy and requirement traceability.**
- **Built a data-driven generator expanding a YAML catalogue of 270+ state transitions, 38 states and 33 events into roughly 290 executable tests, removing hand-written duplication.**
- **Engineered the bench device layer so each instrument is owned by a single thread against a shared monotonic clock, covering a pyvisa/USBTMC/SCPI programmable AC source, CAN serial-bus monitoring and Modbus relay and probe I/O.**

Software Developer (C++)

Thermo Fisher Scientific (via Ixperta (contractor)) · 2023 – 2024

Electron microscopy software for high-resolution detector hardware operation and image acquisition. Built server-side C++ applications, FPGA-based data processing pipelines with gRPC networking, and backend logic for a C#/.NET WPF image processing and display application. Worked in an international Agile team with bi-weekly stakeholder demos.

- **Implemented server-side software in C++, including hardware abstraction layers (HAL) and inter-process communication (IPC) using COM and MIDL on Windows.**
- **Developed embedded software for FPGA hardware, using gRPC network communication in C++, Python and Linux environments.**
- **Optimized the data acquisition pipeline and implemented robust error handling.**

Software Engineer II

Honeywell · 2022 – 2023

Onboard Maintenance System that monitors the condition of aircraft and diagnoses issues during flight. Architected and implemented the DiagTest framework, a state machine-based diagnostic test engine in C++ managing 15+ operational states for aircraft member system testing during flight and maintenance operations.

- **Designed a job-based execution system with DiagTestJob and DiagTestMsgQueProc for asynchronous test orchestration, implementing message queue processing patterns for reliable real-time diagnostics.**
- **Engineered a test interference management system preventing conflicting diagnostic tests from running simultaneously, ensuring aircraft system safety and data integrity.**
- **Implemented four diagnostic test modes (data collection, passive monitoring, interactive, non-interactive) with timeout management, precondition handling and aviation-standard command/response protocols.**

Application Developer

Trusted Network Solutions · 2020 – 2022

Kernun Adaptive Firewall. Implemented a forward proxy in C++ on Linux featuring HTTP/HTTPS proxying with TLS/SSL inspection, OSCP integration for certificate validation, Kerberos authentication and LDAP user/group lookup, plus the ipmon network monitoring daemon driving nftables firewall rules from real-time IPv4/IPv6 address changes.

- **Built comprehensive proxy infrastructure with certificate generation and management (CA, client certificates), a dynamic configuration builder/parser system, and a manager/partition architecture for modular daemon design.**
- **Integrated security components including Suricata (intrusion detection) and nftables (static firewall), aggregating logs and managing connection information in PostgreSQL.**
- **Implemented real-time network interface monitoring in ipmon using NETLINK sockets, handling both IPv4 and IPv6 address management, with Unix domain sockets for IPC and dynamic nftables rule management.**

Independent IT Contractor

Self-employed

Independent contracting for business and consumer clients: CI/CD automation pipelines and Linux server administration for B2B customers, and IT support, consumer hardware retail and security consulting for B2C customers.

- **B2B: CI/CD automation pipelines and Linux server administration.**
- B2C: IT support freelancing, consumer hardware retail and security consulting.

Automation Developer & IT Systems Specialist

IBM · 2014 – 2016

Enterprise IT operations: infrastructure monitoring, incident management and an automation solution that replaced manual ticketing and server administration work.

- Developed an automation application that replaced manual workload in ticketing operations and basic server administration, using Bash, PowerShell and JavaScript for cross-platform server management and applying automata theory to design state-driven workflows.
- **Monitored Unix/Windows servers and resolved production incidents, working hands-on with Active Directory and VMware virtualization.**

Skills

Embedded & RTOS

C++ (C++17/C++20)	Senior
Embedded software / ARM / RTOS	Medior
State machine design patterns	Senior
Hardware abstraction layers & IPC (COM/MIDL, Unix sockets)	Medior
FPGA-adjacent development & gRPC	Medior
Flatbuffers (binary serialization)	Medior
Qt/QML	Working knowledge

Networking & Security

Linux	Senior
Linux networking, TCP/IP, Netlink, nftables	Senior
Encrypted traffic inspection (TLS 1.3, OSCP)	Medior
Kerberos/Heimdal, LDAP, PostgreSQL, Suricata IDS	Medior
Forward/TCP/HTTP proxy design	Senior
Rust	Working knowledge

Tooling

Git	
Bash	
Docker / Podman	
Conan (C/C++ dependency management)	
CI/CD (Jenkins, TeamCity, GitLab CI, GitHub Actions)	
PowerShell	
Enterprise Architect (UML)	
GDB, Wireshark, VirtualBox	

Verification & HIL

Python	Medior
Hardware-in-the-loop test design	Senior
pyvisa / USBTMC / SCPI instrument control	Medior
Modbus I/O cards & CAN serial-bus diagnostics	Medior
Safety-certified product verification (UL/CSA, ANSI Z21.20, NFPA 54)	Medior
Requirement traceability & test evidence reporting (openpyxl, HTML, Mermaid)	Medior
C# / .NET	Medior
Perl test frameworks	Medior

AI / LLM / MCP

Model Context Protocol server development	Senior
Agentic workflow design & governance (agents, skills, policy hooks)	Medior
Prompt engineering (RAG, few-shot, prompt caching, structured outputs, tool calling)	Medior
LLM provider APIs & SDK integration	Medior
JavaScript / TypeScript	Medior
Node.js	Medior

Open Source & Projects

MCP-Prompts — 97 stars, 21 forks | github.com/sparesparrow/mcp-prompts
TypeScript · Node.js · PostgreSQL + pgvector · Docker · REST + SSE · AWS

rust-network-mgr | github.com/sparesparrow/rust-network-mgr
Rust · Netlink · nftables · epoll · Lock-free data structures

Education & Training

Masaryk University, Faculty of Informatics — Computer Science

Coursework only — no academic title — Coursework in programming, Linux and networking.

Coursera.org / DeepLearning.AI — Machine learning and LLM engineering

Coursework only — no academic title — Deep Learning Specialization (four machine learning courses by Andrew Ng), plus data analysis, prompt engineering and building LLM agents.

Self-directed study — Design patterns, software architecture and offensive security

Coursework only — no academic title — Design Patterns (refactoring.guru) and Hack The Box cybersecurity training; self-studied the Certified Ethical Hacker syllabus (exam not taken).

Professional development reading — Software engineering practice

Coursework only — no academic title — Software Engineering at Google, The Pragmatic Programmer, Clean Architecture.

Languages

Czech (native) · Slovak (native) · English (C1) · German (A1)

References

References available on request — two to three referees and work certificates (Arbeitszeugnisse) can be provided on request.

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